Pre-FEED design gate

DRAFT — simulator screening, not approved for construction.

Verdict: ROBUST PASS across 2 stated operating scenarios.

Operating envelope

Scenario	Load	Result	Governing check	Utilisation	Mass closure	Basis
today	100%	PASS	PV-1767 Fair flood fraction vs limit	78%	0%	current throughput
target	125%	PASS	PV-1767 Fair flood fraction vs limit	98%	0%	case target

Energy closure: not checked: the solver does not report enthalpy on the boundary streams

Discipline impact register

Discipline	Status	Impact	Evidence
Process	pass	design reaches 125 % (max 127 %, re-solved now)	DWSIM re-solve of the combined design on the current model
Mechanical	not affected	no new physical equipment	proposal calculation and provisional ratings
Electrical	not affected	no new driver identified	constructability gate
Piping Utilities	open	no calculation adapter: tie-ins, line sizing, supports and utility headers remain open for PV-1767	proposal topology (no engine for this discipline)
Controls	open	no calculation adapter: control narrative, split/load-sharing control, cause & effect and permissives remain open for PV-1767	proposal topology (no engine for this discipline)
Process Safety	open	no calculation adapter: change-focused HAZOP, relief case and isolation for the changed equipment remain open for PV-1767	proposal topology (no engine for this discipline)
Civil Structural	open	no calculation adapter: plot, access, lifting and foundations remain open for PV-1767	proposal topology (no engine for this discipline)
Robustness	fail	draft (assumed values): operating envelope 1/2 scenarios pass (fails: Target, feed +10 %)	DWSIM solves of the pre-FEED operating envelope

Decision-changing evidence

No decision-changing field input was identified from the recorded changes.

Change portfolio

Change	Role	Verified	Capacity	Cost class	New equipment
Retray PV-1767 with high-capacity trays	standalone target	yes	127.5%	not stated	—
Combined proposed design	combined envelope	yes	125.0%	not stated	—

not ranked: sourced numeric cost, outage and energy values are required

Preliminary equipment selection

No new physical equipment in the selected design.

Critical-loop dynamics

Not run: no measured dynamic model.

Needed: equipment liquid/vapour holdup; controller action and tuning; valve/drive response; step-test time constant and dead time; trip and permissive logic.

Change-focused HAZOP and relief screen

Tag	Deviation	Cause	Consequence	Existing safeguard	Action

Cost and constructability gate

Cost classes on the proposals: not stated.

Outage required: no identified outage work. Exact tie-ins: 0.

No sourced numerical estimate supplied; these items remain unpriced: installed equipment; piping/tie-ins; electrical and controls; civil/structural; outage production loss.

Commissioning acceptance

Status: **OPEN** against scenario target.

No solved acceptance check maps to the changed equipment; define the acceptance measurements before issue.

Gate and limitations

- A pass means all stated simulator checks passed in all stated scenarios; it is not mechanical or vendor approval.
- New equipment remains provisional until guaranteed curves and materials are checked.
- The automated hazard register does not replace a multidisciplinary HAZOP or relief-system design.
- Missing inputs stay open; PlantAgent does not invent curves, dynamic constants or costs.

Economics and uncertainty

Economics (ILLUSTRATIVE prices; order of magnitude)

- draft (assumed values): Capex ≈ $0k (class 5: $0k–$0k) plus unpriced items (replace PV-1767): capex is a lower bound; product price not given: margin, payback and NPV open (energy and utilities $0k/yr). All prices ILLUSTRATIVE.

Proposal	Stated class	Factored capex	Check
Retray PV-1767 with high-capacity trays		$0 + to price: replace PV-1767	

- Added Process feed: 552 t/d × 330 d; product margin $0/yr (solved 1.250 × base, capped at the 125% target; energy from dwsim solves (design at 125% vs today at 100 %)).
- Outage for tie-ins/cleaning: 2 d → $0 (muse_default shutdown for a tie-in or exchanger clean).
- Payback None years; NPV $0 (capex band: $0 to $0).

Uncertainty of the design (Monte Carlo over assumed and illustrative inputs)

- Target 125% met in 92% of 48 samples; max throughput P10/P50/P90 127% / 127% / 128%; most often limited by PV-1767: Fair flood fraction vs limit (100%); biggest driver feeds.FEED.P_bar (assumed).
- Method: Latin hypercube, 48 samples (seed 20261101), uniform ± band per input kind; each sample re-solved (dwsim): capacity bisection plus the solve at the target; 48 of 48 samples solved.

Input	Kind	Base	± band
units.PV-1767.dP_bar	assumed	1.703	10 %
feeds.FEED.P_bar	assumed	25.5	10 %
feeds.FEED.T_C	assumed	-9.5	10 %

- PV-1767: Fair flood fraction vs limit at the target: load P10/P50/P90 98% / 98% / 99%.